Worksheet 3 - 14.1, 14.3-5

Complete in Discussion

1. Find the tangent plane to
$$f(x,y) = \sqrt{x^2+y^2}$$
at $(x,y) = (6,8)$. Use the tangent plane to approximate $\sqrt{130}$ by evaluating the tangent plane at $(x,y) = (7,9)$. How does this compare to the actual value? Illustrate with a diagram. (Screenshot of Geogebra recommended).

2. Match the figures below with the equations:

   (a) $z = \sin(x)\sin(y)$

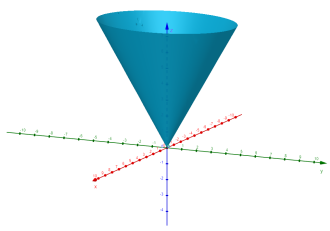
   (b) $x^2 - y + z^2 = 0$

   (c) $z = \sqrt{6x^2+3y^2}$

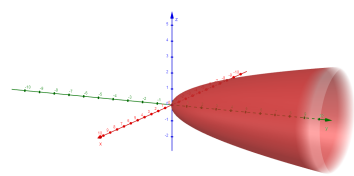
   (d) $z = \ln(x^2+y^2)$

f(6,8) = sqrt(6^2+8^2)
= sqrt(100)=10
fx= 1/2(x^2+y^2)^-1/2*2x
@ (7,9) = 1/2(7^2+9^2)^-1/2*2*7
= .6139
fy = 1/2(x^2+y^2)^-1/2*2y
@ (7,9) = 1/2(7^2+9^2)^-1/2*2*9
= .7894
z = z0+fx(x-x0)+fy(y-y0)
z = 10 + .6139(7-6)+.7894(9-8)
z = 11.4033

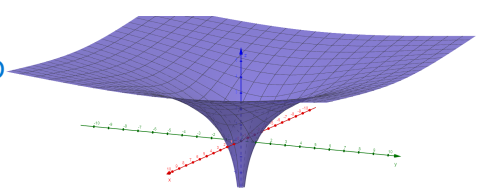
I: C



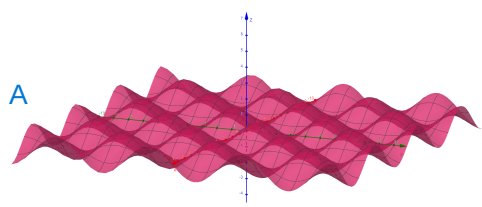
II: B



III: D



IV: A



3. Verify Clairaut's Theorem on
$$f(x,y) = \frac{4}{x^2+y^2+2}$$
by showing $f_{xy} = f_{yx}$.

f = 4(x^2+y^2+2)^-1/2
fx = -2(x^2+y^2+2)^-3/2*2x
fx = -4x(x^2+y^2+2)^-3/2
fxy= 12xy(x^2+y^2+2)^-5/2

fy = -2(x^2+y^2+2)^-3/2*2y
fy = -4y(x^2+y^2+2)^-3/2
fyx = 12xy(x^2+y^2+2)^-5/2

fxy = fyx

Extra Quiz/Exam Practice

4. Sketch the domain of the function

$$f(x,y) = \frac{3x - \sqrt{y}}{\sqrt{x^2 - y^2}}.$$

5. Find the point(s) (a,b) where the tangent plane to

$$f(x,y) = 2 - x^4 + 2x^2 - y^2 + 3.$$

at (a,b) is horizontal.

6. Use the chain rule to find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$ for

$$z = x^2 + 2xy + y^3$$

where $x = s + t$ and $y = 2st$. Then evaluate these directly (without using chain rule) by first plugging in the functions values of x and y into z and then taking the standard derivative.

Challenge Questions

7. Let

$$f(x,y) = \frac{x^2y + 2e^{xy}}{\ln(x)}.$$

Find $f_x(e,0)$ two ways. First calculate $f_x(x,y)$ using the standard partial derivative techniques and then plug in the point at the end. As a second approach, evaluate $f(x,0)$ to reduce f to a function of one variable. Then take the one variable derivative.

8. In Exercise 2 above, for what values of x and y does this tangent plane approximation give an exact value?

9. Use the tangent plane to the surface $f(x,y) = \sin(xy)$ at the point $(1,\pi)$ to show that $\sin(3) \approx \pi - 3$. [Note: This was an Exam 1 question in Fall 2021]

7)
f = (x^2y+2e^xy)/(ln(x))
fx = (ln(x)*(2yx+2ye^xy)-1/x*(x^2y+2e^xy))/(ln(x))^2
fx(e,0) = (ln(e)*(2*0*e+2*0e^e*0)-1/e*(e^2*0+2e^e*0))/(ln(e))^2
= -1/e*(1+2) = -2/e
f(x,0) = (x^2*0+2e^x*0)/(ln(x)) = 2/ln(x)
f' = -2*(1/x)/(ln(x))^2
f'(e) = -2*(1/e)/(ln(e))^2 = -2/e

9)
f(1,pi) = sin(1*pi) = 0
fx = y*cos(xy) @ (1,pi) = pi * cos(1*pi) = -pi
fy = x*cos(xy) @ (1,pi) = 1* cos(1*pi) = -1
z = -pi (x-1) -1 (y-pi)
point = (1,3)
z = -pi(1-1)-1(3-pi) = pi - 3

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